

Testing & Quality Assurance Plan

WikiMove | Student: Sanchita Suryawanshi (Roll No: 150096724115) | Academic Portfolio 2026

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Project WikiMove – Migrating a Company Knowledge Base to a New Platform

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Student Name: Sanchita Suryawanshi
Roll Number: 150096724115
Institution: [Department of Computer Science & Engineering / University Placeholder]
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1. Executive Summary & QA Objectives

This Testing and Quality Assurance document establishes the verification and validation framework for Project WikiMove. Because WikiMove is an academic software engineering proposed system, this document defines planned test specifications, test cases, boundary value analysis models, decision tables, and quality metrics.

Academic Testing Notice: In strict compliance with academic instructions, no fake executed test results are presented. All test cases are documented with:

- Status: Planned
- Actual Result: To be executed during implementation

The primary QA objectives are:

- 1. Validate 100% correct triage classification of legacy pages.
- 2. Certify that zero broken internal links exist across migrated pages (NFR-01).
- 3. Ensure search index discoverability for all 7,000 kept pages (NFR-02).
- 4. Verify mandatory content ownership binding (Zero Orphan Policy).

2. Testing Scope & Strategy

2.1 Testing Scope

- In-Scope: Page inventory indexing, triage state transitions, mandatory owner validation, migration batch progression, URL rewrite verification, link crawling, search indexing, and 90-day governance reminder triggers.
- Out-of-Scope: Stress testing external commercial cloud vendors, live load testing corporate network gateways, automated natural language translation quality.

2.2 Testing Levels

- Component Verification: Validation of data models, regex link parsers, and metadata schemas.
- Integration Verification: Interaction between inventory catalog, triage decision engine, and owner registry.

- System Verification: End-to-end evaluation of page lifecycle from discovery to validated migration.
- User Acceptance Verification (UAT): Review workflow trials with sample engineering team members.

3. Functional Test Cases

Test ID	Requirement ID	Test Title	Test Procedure	Expected Result	Status	Actual Result
TC-01	FR-01	Ingest Legacy 14k Inventory	Import legacy dump of 14,000 pages with metadata.	14,000 pages indexed with correct creation and modified dates.	Planned	To be executed during implementation
TC-02	FR-02	Page Review Content Display	Select page WP-01042 and open inspection screen.	Renders HTML/Markdown preview, legacy URL, author, and revision history.	Planned	To be executed during implementation
TC-03	FR-03	Triage Action Selection	Select Keep, Archive, or Delete and submit note.	State transition recorded with reviewer timestamp.	Planned	To be executed during implementation
TC-04	FR-04	Keep Page Workflow & 50% Cap	Mark page as KEEP and verify batch placement.	Transitions to KeepPendingMigration; cumulative kept tracks ~7,000.	Planned	To be executed during implementation
TC-05	FR-05	Archive Page Workflow	Mark page as ARCHIVE with retention note.	Generates read-only archival stub; removes from migration queue.	Planned	To be executed during implementation
TC-06	FR-06	Delete Page Grace Period	Mark page as DELETE.	Applies 14-day soft-delete grace window before purge.	Planned	To be executed during implementation
TC-07	FR-07	Zero Orphan Owner Enforcement	Attempt to migrate kept page without assigning owner.	System blocks migration and displays mandatory owner error.	Planned	To be executed during implementation
TC-08	FR-08	Migration Batch Scheduling	Group 500 kept pages into BATCH-01.	Batch allocated at 25 pages/person-day velocity benchmark.	Planned	To be executed during implementation
TC-09	FR-09	Migration Status Transition	Execute migration on BATCH-01.	Status updates from Queued to Migrated and Validation Pending.	Planned	To be executed during implementation
TC-10	FR-10	Link Integrity Crawler (NFR-01)	Crawl 100 internal intra-wiki links in migrated batch.	All rewritten URLs resolve to HTTP 200 (0.0% broken links).	Planned	To be executed during implementation
TC-11	FR-11	Search Discovery QA (NFR-02)	Query top 20 domain keywords against target platform.	All relevant migrated pages return within top 5 search hits (< 1.0s).	Planned	To be executed during implementation
TC-12	FR-12	Schedule Gap Calculation	Check progress burn-down against 39-week licence.	System correctly alerts +3.0 week schedule overrun conflict.	Planned	To be executed during implementation

TC-13	FR-13	Risk Exposure Ranking	Compute risk exposure \$E = P \times I\$ for R1..R4.	R2 ranked #1 (\$E=4.2\$), R1 ranked #2 (\$E=3.6\$), R3 #3, R4 #4.	Planned	To be executed during implementation
TC-14	FR-14	Management Reporting View	Generate executive summary on Screen 1 Dashboard.	Displays 14k total, 7k kept, 350 p-d triage, 280 p-d migration.	Planned	To be executed during implementation
TC-15	FR-15	90-Day Governance Reminder	Simulate 91 days elapsed since last review.	System triggers owner revalidation prompt and flags in Screen 5.	Planned	To be executed during implementation

4. Boundary Value Analysis (BVA)

Boundary Value Analysis tests critical edge conditions at input and capacity thresholds:

Parameter / Dimension	Valid Range	Min Boundary	Just Above Min	Nominal	Just Below Max	Max Boundary	Out of Bounds
Page Inventory Ingestion	1 to 14,000	1	2	7,000	13,999	14,000	0, 14,001
Daily Triage Quota	1 to 40 p/d	1	2	20	39	40	0, 41
Migration Batch Size	1 to 500 pg	1	2	250	499	500	0, 501
Search Response Latency	0 to 1.00s	0.01s	0.10s	0.50s	0.99s	1.00s (Limit)	> 1.00s (NFR Fail)
Link Integrity Ratio	0.0% broken	0.0% (Pass)	0.1%	0.0%	0.0%	0.0%	> 0.0% Broken

5. Decision Tables

5.1 Page Triage Decision Table

Guides reviewers in classifying legacy content:

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Is content technically current and accurate?	Y	Y	N	N	N
Does content have active business relevance?	Y	Y	N	N	N
Is named content owner available?	Y	N	—	—	—
Is content required for compliance / legal audit?	—	—	Y	N	N
Is content empty, duplicate, or transient notes?	N	N	N	Y	N
Actions					
Action: KEEP (Queue for Migration)	X				
Action: Escalate for Owner Designation		X			
Action: ARCHIVE (Cold Storage Snapshot)			X		
Action: DELETE (14-Day Grace Window)				X	
Action: Flag for Technical Writer Triage					X

5.2 Migration QA Certification Decision Table
Validates migrated pages before production sign-off:

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
HTML / Markdown parsed without syntax error?	Y	Y	Y	N
All intra-wiki hyperlinks resolve to HTTP 200?	Y	N	Y	—
Page discoverable in target search index?	Y	Y	N	—
Named content owner verified in directory?	Y	Y	Y	—
Actions				
Action: Certify Page on New Platform	X			
Action: Log Link Defect & Remap URL		X		
Action: Trigger Search Re-Indexing			X	

Action: Reject Batch & Flag Parsing Defect				X
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6. Defect Removal Efficiency (DRE)

6.1 Mathematical Formulation

Defect Removal Efficiency measures the efficacy of quality verification activities in identifying and eliminating defects prior to final production cutover:

$$\text{DRE} = \frac{E}{E + D} \times 100$$

Where:

- E = \text{Number of defects discovered and resolved before release (pre-cutover QA)}
- D = \text{Number of defects discovered after release (post-cutover by end users)}

6.2 Planned Quality Measurement Approach

- In the WikiMove migration pipeline, E encompasses broken hyperlinks, search indexing failures, and unassigned owner defects identified by automated crawlers during Phase 7 testing.
- Target project goal: Achieve $\text{DRE} \geq 95.0\%$ before legacy platform shutdown.

Illustrative Example
— Planned Quality Benchmark Model:

If pre-release

crawlers detect and fix 190 broken links (\$E = 190\$) and end users report 10 missing page redirects post-migration (\$D = 10\$):

$$\text{DRE} = \frac{190}{190 + 10} \times 100 = \frac{190}{200} \times 100 = \mathbf{95.0\%}$$

7. Defect Density

7.1 Mathematical Formulation

Defect Density measures the normalized frequency of defects per unit size of tested content:

$$\text{Defect Density} = \frac{\text{Total Number of Valid Defects Discovered}}{\text{Total Volume of Tested Content}}$$

7.2 Measurement Approach

In WikiMove, content size is measured in units of 1,000 migrated pages (KPages):

$$\text{Defect Density} = \frac{\text{Valid QA Defects}}{\text{Total Kept Pages Migrated (in KPages)}}$$

Illustrative Example — Planned Quality Benchmark Model:

For the total migrated scope of 7,000 kept pages (7.0 KPages), if QA testing uncovers a total of 35 defects across link crawling, search indexing, and formatting:

$$\text{Defect Density} = \frac{35 \text{ defects}}{7.0 \text{ KPages}} = \mathbf{5.0 \text{ defect s / 1,000 pages}}$$

$$(0.005 \text{ defects / page})$$

Target ceiling: $\leq 8.0 \text{ defects per 1,000 pages}$ prior to final sign-off.

8. Test Completion & Certification Criteria

A migration batch is certified for production cutover when:

1. 100% of planned functional test cases (TC-01 through TC-15) execute successfully.
2. 0.0% broken internal hyperlinks remain within the migrated scope (NFR-01).
3. 100% of migrated pages return in search index queries within < 1.0 second (NFR-02).
4. 100% of migrated pages have a confirmed named owner (Zero Orphan Policy).
5. Defect Removal Efficiency (DRE) meets or exceeds the 95.0% threshold.